

Online Supplement

Concentration, Not Job Loss: Tax–Benefit Cushioning of Earnings Shocks in the Universal Credit Era

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This online supplement collects the exploratory and benchmark material referenced from the main paper: the elasticity/pass-through grid, the reallocation sensitivity, the observed-outturn stress scenario, the HMRC destination-panel diagnostic, the Economic Prosperity Deal counterfactual (withdrawn as a quantitative result), the demographic diagnostic, the supply-chain sensitivity, and the constituency machinery. All numerical values are generated from the pinned artifacts in the replication package; conventions (assignment SDs after $\pm$, exploratory draw counts) follow the main paper’s Monte Carlo appendix.

S1 Elasticity and pass-through grid

The regenerated grid spans $\varepsilon \in \{0.4, 1, 2, 3\}$ and $\pi \in \{0.5, 0.75, 1\}$. Because only five displacement assignments are used per cell, it is an exploratory scaling diagnostic. Equal products $\varepsilon\pi$ generate equal deterministic wage-cut shocks; Bernoulli displacement outcomes vary across assignments. The grid is not an uncertainty interval.

S2 Reallocation sensitivity

The reallocation exercise uses five paired assignments and is exploratory. Under the full schedule, the central 28.3 per cent penalty produces a mean gross loss of £264 million, Exchequer cost of £148 $\pm$ 130 million, cushioning of 41.4 $\pm$ 3.1 per cent, and zero measured BHC-poverty change. With a three-month lag, gross loss rises to £432 million and the Exchequer cost to £244 $\pm$ 215 million. The 14 per cent penalty gives gross loss of £131 million and Exchequer cost of £74 $\pm$ 64 million. These pound levels are not commensurable with the economy-wide wage-cut scenario because only selected movers’ penalty is removed.

S3 Observed-outturn stress scenario

The descriptive May 2025–February 2026 export window falls 12.71 per cent against its two-year same-month baseline. This is not a causal tariff estimate: it includes prices, exchange rates,

front-running, rerouting and other demand conditions.

The downside displacement scenario affects 17.3 thousand weighted workers on average, is cushioned by 36.6 ± 4.9 per cent and costs the Exchequer $\pounds 419 \pm 228$ million. The wage-cut scenario is cushioned by 44.1 per cent and costs $\pounds 517$ million. This is an observed-outturn downside stress case, not a tariff-caused or preferred estimate; the validation artifact also reports signed net exposure before expanding divisions are clipped.

S4 HMRC destination-panel diagnostic

Across 1194 HS4 products, the capped value-weighted specification estimates a -33.9 per cent change in exports to the US relative to the same products sent to the three comparison destinations (95 per cent interval -47.8 to -16.4 per cent, $p = < 0.001$). Results have similar signs against each destination, and May 2019–23 pseudo-policy estimates are individually insignificant. This is not robust evidence of a common product effect: equal product weights give -3.2 per cent ($p = 0.519$), and restricting the sample to 2023 onward gives -21.1 per cent ($p = 0.121$). The estimated differential trend is itself borderline ($p = 0.046$). Most importantly, broad steel and auto chapters fall -22.5 per cent ($p = 0.283$), compared with -34.9 per cent for other products ($p = 0.001$): the tariff-intensity ordering fails. Figure 1 displays the monthly gap. The panel supports a post-policy decline concentrated in large US export products, but it cannot attribute that decline to tariffs or validate the tariff-rate mapping.

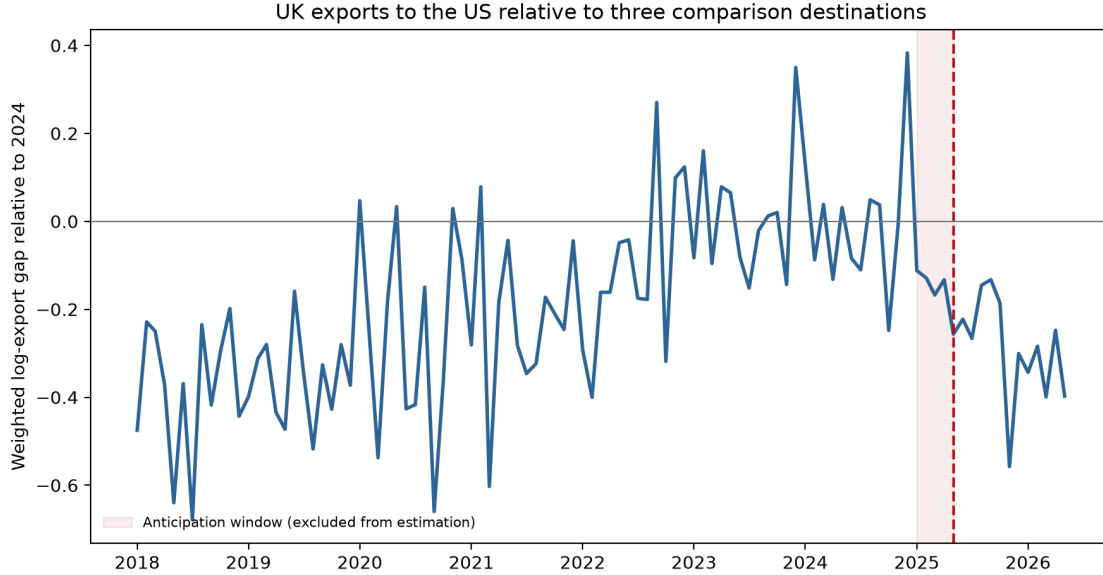


Figure 1: Monthly capped-value-weighted gap between UK HS4 exports to the United States and mean exports of the same products to Canada, Japan and Australia, normalised to calendar 2024.

The machine-readable specification, clustered standard errors, individual destination checks, sample-start sensitivities, tariff-intensity groups and May 2019–23 pseudo-policy estimates are stored in the event-study JSON artifact; the plotted observations are stored in its companion monthly CSV. Pre-policy value weights are capped at the 99th percentile, giving an effective

product count of 107 despite 1194 included HS4 products. Uncapped and equal-weight estimates disclose the remaining concentration sensitivity.

S5 Economic Prosperity Deal counterfactual (withdrawn as a quantitative result)

On paired seeds, full minus EPD displacement differs by 3.7 ± 2.8 thousand workers, $\pounds 204 \pm 192$ million of gross earnings and $\pounds 96 \pm 101$ million of annual Exchequer cost. The contrast is driven primarily by the lower auto tariff and the exemption of pharmaceuticals in the EPD schedule, and it should be interpreted as the fiscal value of changing the declared sector shock while holding the household model and assignment draws fixed. Two caveats withdraw it as a quantitative result. First, every paired difference is comparable to or smaller than its own assignment dispersion—the Exchequer contrast of $\pounds 96$ million carries an assignment SD of $\pounds 101$ million—so at the reported draw count the EPD contrast is not distinguishable from allocation noise. Second, it prices a change in the sector calibration whose own rank-order check fails; under the EPD schedule the top-ten rank correlation is -0.62. The comparison is therefore retained as a qualitative illustration of the counterfactual machinery, not as an estimate of the deal’s value. It also does not include quota allocation, above-quota trade, price responses, NHS pricing concessions, or other general-equilibrium effects. Figure 2 summarises the paired comparison. Autos drive much of the difference; the scenario applies the in-quota rate to fixed exposure without modelling quota allocation or future above-quota trade.

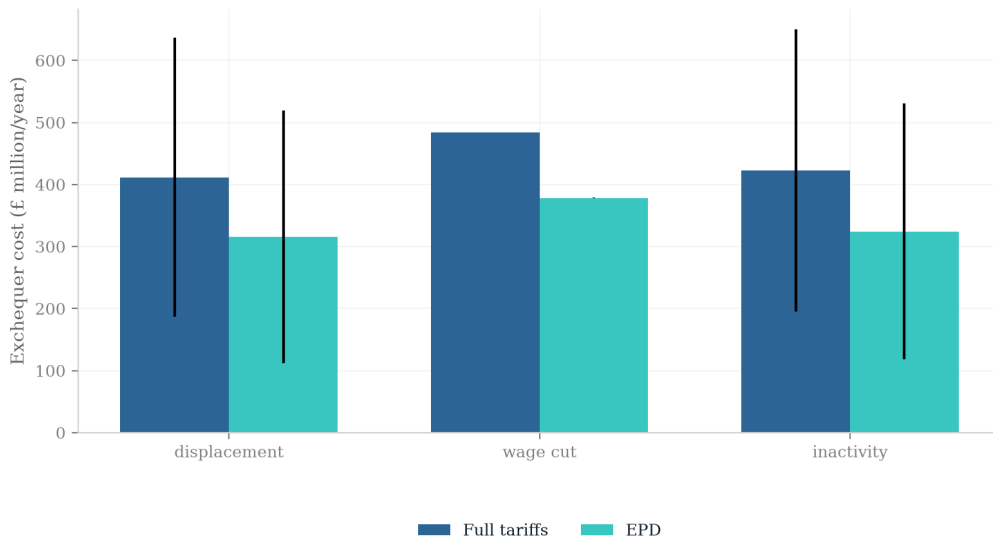


Figure 2: Full-schedule and EPD scenario comparison. Error bars are assignment standard deviations. The paired differences are comparable to or smaller than their assignment dispersion; no quantitative value should be placed on them.

S6 Demographic diagnostic

The gender, age and household-type breakdown uses five Bernoulli assignments and is exploratory. Selected exposed-sector workers are disproportionately men and concentrated in prime and older working ages, but individual subgroup estimates are too assignment-sensitive for precise rankings. The demographic artifact is therefore a diagnostic rather than an inferential table.

S7 Supply-chain sensitivity

The input–output scenario adds upstream requirements using the ONS domestic Leontief inverse. The accounting shock has a 1.95 direct-plus-upstream factor. It is not a lower bound and is not multiplied by local-employment multipliers. In the exploratory ten-assignment microsimulation, displacement affects 29.7 thousand weighted workers on average, costs the Exchequer $\pounds 682 \pm 313$ million and raises relative BHC poverty by 0.028 percentage points. The wage-cut variant costs $\pounds 955$ million and leaves measured relative poverty unchanged. Both runs predate the Universal Credit award-cache correction—a 2.8-point offset where it has been measured on the displacement margin—so their levels are not comparable with the main paper’s bounds table; they are retained as an exploratory extension. These figures are accounting sensitivities, not all-channel causal estimates.

S8 Constituency machinery

The enhanced-FRS constituency layer uses imputed industries rather than observed local industry structure. Its outputs are averaged over 20 paired worker-assignment draws (seeds 0–19), with each household’s disposable-income change divided by household size before constituency means are constructed; the CSV artifact reports both the draw mean and the assignment-draw standard deviation. Because the constituency weights contain no observed local industry mix and the SIC imputation is held fixed, the outputs are synthetic demographic and income projections, not constituency exposure estimates. No map or seat-level figures are reported in the main paper; the machinery is documented here so that future work can populate it with observed local industry data.